

VIANKA VANESSA CASTRO ORDOÑEZ

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EDUCATION

Universidad del Valle de Guatemala

B.S. in Computer Science and Information Technology; 8th semester

Guatemala City, Guatemala

Jan. 2023 - Present

Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Databases, Software Engineering, Artificial Intelligence, Operating Systems, Automata and Compiler Theory.

Colegio Decroly Americano

Bachelor of Science and Letters; High Honors recognition

Guatemala City, Guatemala

2011 - 2022

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, PHP, Kotlin, C, SQL, HTML, CSS, Rust

Frameworks & Web: Next.js, React, Laravel, Vue.js, Node.js, Vite, REST APIs, MVC

Databases & Cloud: PostgreSQL, SQL Server, SQLite, Firebase Firestore, AWS and Azure fundamentals

Tools & Practices: Git, GitHub, Docker, Docker Compose, Jira/Atlassian, CI/CD fundamentals, Agile/Scrum, Visual Studio, IntelliJ, Android Studio, LaTeX, MOS Excel

Languages: Spanish: Native; English: Upper-intermediate (B2)

PROFESSIONAL EXPERIENCE

Computer Science Teaching Assistant

Universidad del Valle de Guatemala

Jun. 2025 - Present

- Support students in Object-Oriented Programming and Algorithms/Dynamic Programming through practice sessions, debugging guidance, and code review.
- Explain classes, inheritance, modular design, algorithmic thinking, recurrence, and dynamic programming in clear, practical terms.
- Review assignments and provide structured feedback on correctness, readability, and problem-solving approach.

Mathematics Teaching Assistant

Universidad del Valle de Guatemala

Jan. 2024

- Tutor university students in logic, set theory, relations, combinatorics, probability distributions, expected value, and structured problem solving.
- Prepare guided exercises and review sessions that translate abstract mathematical concepts into clear, step-by-step solutions.

SELECTED PROJECTS

NekoFit - Nutrition and Wellness in One Place

Technologies: Next.js, React, TypeScript, JavaScript, CSS, REST APIs, Web Storage

[GitHub](#)

- Designed and developed a personal web application to centralize nutrition planning, macro and exchange tracking, hydration, workouts, body progress, shopping, and outfits.

Web Portfolio - Desktop OS

Technologies: React, Vite, JavaScript, CSS, HTML5, ESLint

[Live](#) | [GitHub](#)

- Developed a personal portfolio with a retro desktop OS interface, draggable and resizable windows, dark/light themes, responsive design, a contact form, and a bilingual resume.

Sol Store - E-commerce Web Application

Technologies: Laravel, Vue.js, PHP, PostgreSQL, Docker, Docker Compose, JavaScript

[GitHub](#)

- Developed a full-stack online store for personalized 3D-printed products, including frontend, backend, database, and a containerized development environment.

Medical Robot Decision-Making Simulation

Technologies: Python, Jupyter Notebook, Value Iteration, Q-Learning, GridWorld

[GitHub](#)

- Compared model-based and reinforcement-learning approaches for a stochastic hospital robot using shared states, actions, rewards, and performance metrics.

Lexer Generator

Technologies: Rust, Automata Theory, Regular Expressions, DFA/NFA, Transition Tables

[GitHub](#)

- Built a lexical analyzer generator from YALex specifications using regex parsing, Thompson NFA construction, DFA conversion, minimization, and simulation.

TUTO! - Tutoring Management App

Technologies: Kotlin, Android Studio, Firebase Authentication, Firebase Firestore, Jetpack Compose

[GitHub](#)

- Developed an Android app for individual and group tutoring management, including authentication, scheduling, user profiles, dark/light mode, and Firestore persistence.

LEADERSHIP & INVOLVEMENT

- Recognized as a Distinguished Student at Universidad del Valle de Guatemala for academic performance.
- Strengthened communication, leadership, and teamwork skills through collaborative software projects and academic tutoring.